GRADUATION PROJECT · WIFI CSI SENSING

DP_NLOS Dataset Analysis

Through-wall presence detection with room-level localization

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01 · SCOPE

Dataset Family and Scope Decision

Dataset	Scenario	Task	Classes	Samples	Paper accuracy
DP_NLOS	**Through-wall (5 rooms)**	**Presence + zone**	**6: no presence, rooms 1-5**	**384**	**97.89**
DP_LOS	Open hallway, no walls	Presence + zone	6: no presence, zones 1-5	392	98.46%
DA_NLOS	Through-wall	Activity	3: no activity, walking, walking + arm-waving	384	88.16%
DA_LOS	Open hallway	Activity	3: same as above	392	97.43%

DP, not DA

DP labels encode presence and room (0 = no presence, 1-5 = rooms). DA classifies walking vs. walking + arm-waving; DA_NLOS and DA_LOS were reviewed but excluded.

NLOS, not LOS

Through-wall sensing is the purpose of the project. In LOS the same five positions have no walls between them.

Selected: DP_NLOS

Through-wall presence detection with room-level localization. Our CNN is already trained on this subset.

How DP_NLOS Was Recorded

Hardware

Two ESP32-S3 boards (transmitter and receiver) with directional biquad antennas. The built-in antenna could not connect across the five rooms.

Signal

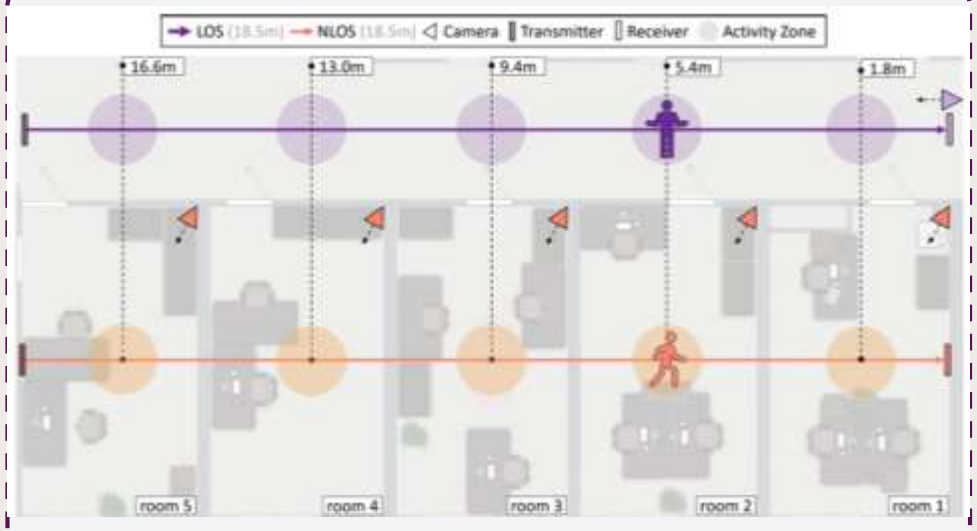
100 packets per second; each carries CSI, the amplitude and phase of the WiFi subcarriers after crossing the walls and the person.

Place

Office hallway with five rooms of about 3.5 × 6.0 m, separated by 25 cm brick walls. Transmitter at the outer wall of room 5, receiver at room 1 (18.5 m apart).

People

One person at a time in one of five fixed zones, one per room: about 2 min walking and 2 min walking with arm-waving per zone, plus about 5 min of nobody present. A camera helped check and clean the recordings.



Top view, NLOS layout (TX/RX 18.5m apart, 5 rooms, 25cm brick walls)

From Raw Signal to Samples

01

Cleaning

A Hampel filter removes outlier spikes from the raw CSI recordings.

02

Windowing

Each recording is cut into 4-second pieces: 400 packets at 100 Hz.

03

Spectrogram

Each piece becomes a 400 × 52 image: time by subcarriers, with colour showing signal amplitude.

Each spectrogram is one sample. With nobody present the pattern looks uniform; when someone walks or waves, visible disturbances appear.

Hampel Filter: Removing Outlier Spikes

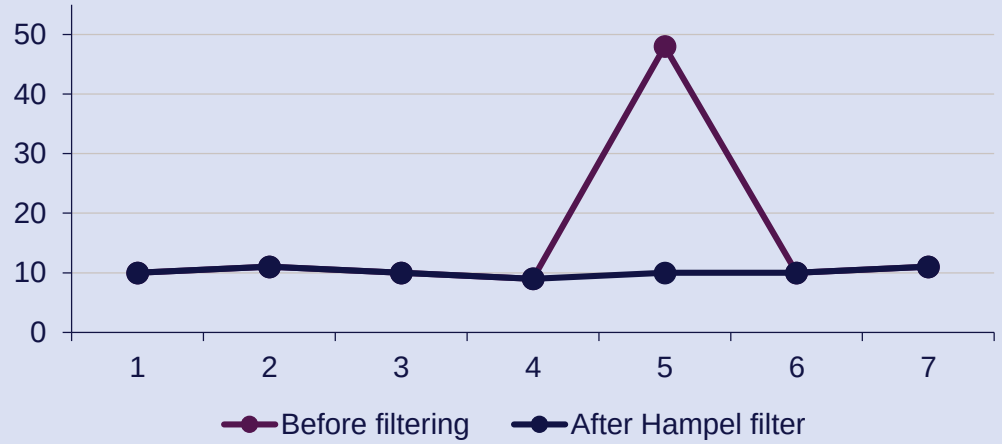
01 Take one subcarrier over time and slide a window of neighbouring packets along it (we start with 11 packets: 5 on each side).

02 In each window, find the median and the MAD (median absolute deviation), a robust measure of spread.

03 Flag a value as an outlier if it lies more than 3 scaled MADs (MAD × 1.4826) from the median.

04 Replace each outlier with the window median and keep every other value unchanged.

Example: 7 packets of one subcarrier



Why the median, not the mean? The spike of 48 pulls the window mean up to 15.6 and hides itself; the median stays at 10, so the spike stands out and is replaced.

Windowing: Cutting the Recording into Samples

W1	W2	W3	W4	...	W29	W30
400 packets 4 s	400 packets 4 s	400 packets 4 s	400 packets 4 s		400 packets 4 s	400 packets 4 s

One 2-minute session = 12,000 packets = 30 windows. Each window becomes one 400 × 52 spectrogram: one sample, labelled with the session's class.

What it is

Windowing cuts a long, continuous recording into fixed-length pieces. Each piece is one sample for the CNN.

Why we need it

The CNN needs a fixed input size, and one recording then gives many samples.

Window size

The paper uses 4 s (400 packets at 100 Hz). We keep it so our data matches DP_NLOS.

Watch out: neighbouring windows are nearly identical, so whole sessions must go to **one split only**. Packets left over at the end of a session (fewer than 400) are dropped. Our plan uses no overlap between windows.

Labels and Dataset Numbers

Label	Meaning
0	Nobody present
1	Person in room 1
2	Person in room 2
3	Person in room 3
4	Person in room 4
5	Person in room 5

384
samples in total, one person in the scene

308 / 38 / 38
train / validation / test (8:1:1 split)

≈60 · ≈75
samples per room · samples with nobody present, estimated from recording times; roughly balanced

Structure of the DP_NLOS Dataset

File / folder	Content
Spectrogram images	The 384 samples, one spectrogram image each, named by its sample ID (for example 336).
trainLabels.csv	Labels for the training split: 308 samples.
validationLabels.csv	Labels for the validation split: 38 samples, used to monitor training and select the best model.
testLabels.csv	Labels for the test split: 38 samples, used only for the final evaluation.
meanStd.csv	Mean (0.5772) and standard deviation (0.1606) of the training set, used to normalize images before they enter the CNN.

Label file format

Two columns, no header. Column A is the sample ID, which links the row to its image; column B is the class label (0 = no presence, 1-5 = person in room 1-5). IDs are global across the dataset, so each split lists only its own IDs and no sample appears in more than one split.

Sample Verification: Spectrogram from DP_NLOS

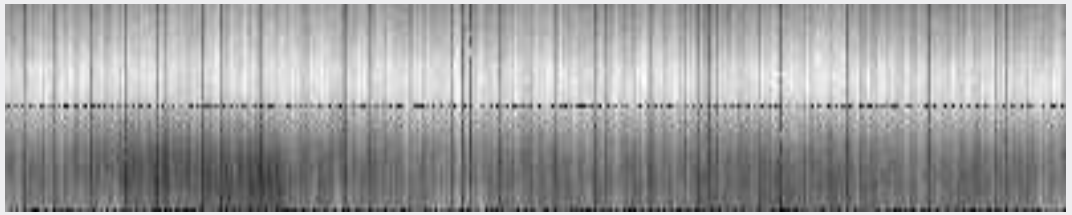


Fig1. sample image from the DP-NLOS dataset

Format verification

- Size and axes match the paper: 400 time steps × 52 subcarriers
- Values are CSI amplitudes

Visual observations

- **Dark vertical lines:** possibly short packet losses or timing gaps; similar lines appear in the paper's Fig. 3
- **Dotted horizontal rows:** subcarriers with very low or constant amplitude, likely unused (not confirmed)
- **Variation along time:** where a person changes the signal; expected weaker for label 0

Visual Comparison: No Presence vs. Presence

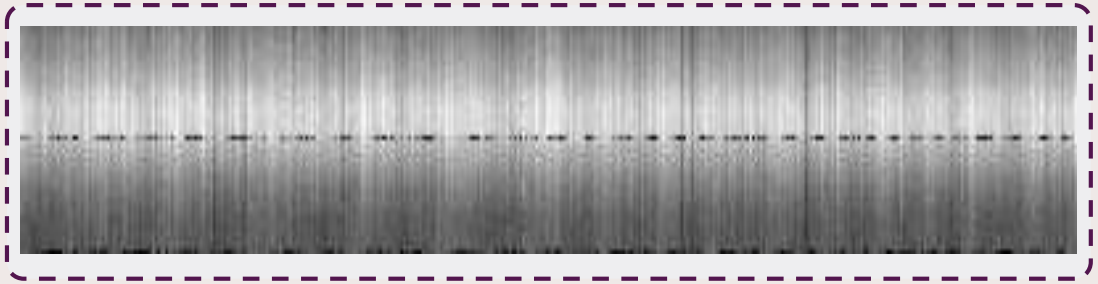


Fig. 2 — No presence, sample ID 196

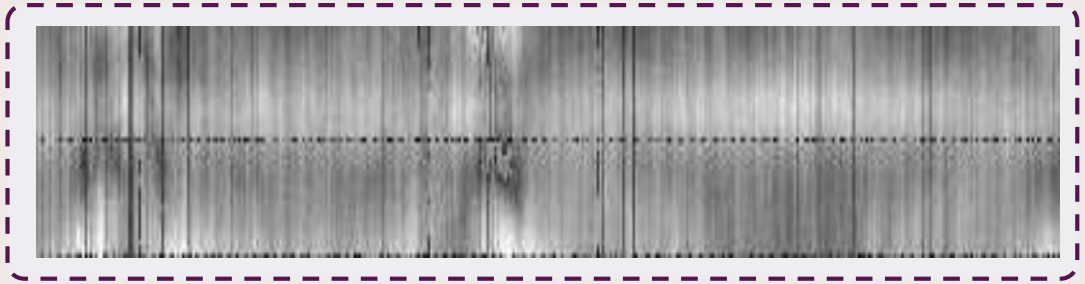


Fig. 3 — Presence in room 5, sample ID 336

Aspect	No presence (label 0)	Presence (labels 1-5)
Channel	Static	Disturbed by a moving person
Amplitude over time	Nearly constant	Fluctuates
Look of the spectrogram	Uniform, steady bands	Irregular patches and variations
What the label tells us	Nobody in the building	Someone present, and in which room

Processing the Collected Data

01

Amplitude

Combine the real and imaginary parts: $\sqrt{(re^2 + im^2)}$.

02

Subcarriers

Keep the 52 L-LTF subcarriers; drop the null ones that stay constant.

03

Hampel filter

Applied per subcarrier along time. The paper gives no parameters; we start with half-window 5 and 3 sigma.

04

Windows

Cut into 400-packet windows (4 s) without overlap, giving 400 × 52 arrays.

05

Save

Store in the DP_NLOS format: match its image size, mode and colour.

06

Label files

Write train, validation and test **CSVs**: sample ID and class, no header.

07

Normalize

Compute mean and standard deviation from our own training windows only.

08

Split by session

Each whole session goes to one split only. Random window splits leak and inflate accuracy.